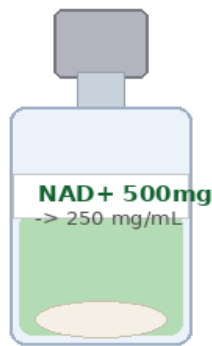


NAD+ (Nicotinamide Adenine Dinucleotide)

Complete Injection Guide — 500mg & 1000mg Vials

Purity 99.8% | Endotoxin-Free | Heavy Metal-Free

What You Have — Two Vial Options



+2mL Sterile Water
250 mg/mL



+2mL Sterile Water
500 mg/mL

Key Diff.

500mg:
250 mg/mL

1000mg:
500 mg/mL

Same 2mL
water both!

Both vials use 2mL sterile water. 500mg gives 250mg/mL. 1000mg gives 500mg/mL. Same dose = different volume. Always check your vial before drawing.

Product Specifications

Parameter	500mg Vial	1000mg Vial
Content	500mg lyophilized NAD+	1000mg lyophilized NAD+
Reconstitution	2mL sterile water	2mL sterile water
Concentration	250 mg/mL	500 mg/mL
Route	IV drip (preferred) / SubQ	IV drip (preferred) / SubQ
Purity	99.8%	99.8%
Endotoxins	Non-detected (<0.1 EU/mg)	Non-detected
Storage (unrecon.)	2-8°C refrigerated	2-8°C refrigerated
Storage (reconstituted)	2-8°C, use within 24 hours	2-8°C, use within 24 hours
Appearance	Clear, slightly yellow solution	Same

SECTION 1 — What is NAD+ and Why Use It?

About NAD+

NAD+ (Nicotinamide Adenine Dinucleotide) is a coenzyme found in every living cell. It is essential for energy metabolism (ATP production), DNA repair, mitochondrial function, and cellular longevity. NAD+ levels decline by up to 50% between ages 40-60, contributing to fatigue, cognitive decline, and accelerated aging. IV or SubQ supplementation rapidly restores cellular NAD+ levels, with significantly higher bioavailability than oral supplementation.

Key Research Benefits

Body System	Research-Supported Effects
Energy & Metabolism	Drives ATP production in mitochondria. Restores cellular energy, reduces fatigue and improves physical performance.
DNA Repair	Activates PARP enzymes and sirtuins (SIRT1-7) that repair DNA damage, protecting against cellular aging and mutation.
Brain & Cognition	Supports neuronal health, improves mental clarity, focus and memory. Research shows potential in Alzheimer's and Parkinson's models.
Anti-Aging (Sirtuins)	Activates longevity proteins (sirtuins) that regulate metabolism, stress resistance, and cellular lifespan.
Cardiovascular	Improves mitochondrial function in cardiac cells, reduces oxidative stress, supports heart health and circulation.
Addiction & Mental Health	Used in clinical IV protocols for addiction recovery, anxiety and depression. Replenishes depleted neurotransmitter precursors.
Inflammation	Regulates NF-kB signalling, reducing chronic inflammation at the cellular level. Supports immune system balance.

Why IV over oral? Oral NAD+ supplements (NR, NMN) must be converted and have ~5-10% bioavailability. IV administration delivers NAD+ directly into circulation at 100% bioavailability, producing immediate cellular effects.

SECTION 2 — Reconstitution Protocol

5-Step Reconstitution Process (Both Vials)



Temp 20°C

Both water
and powder



Sanitise

Alcohol 70%
on cap



Draw water

Exact volume
into syringe



Inject slowly

Along vial
wall only



Verify

Clear solution
no particles

Follow all 5 steps in order. Use exactly 2mL sterile water for BOTH vials.

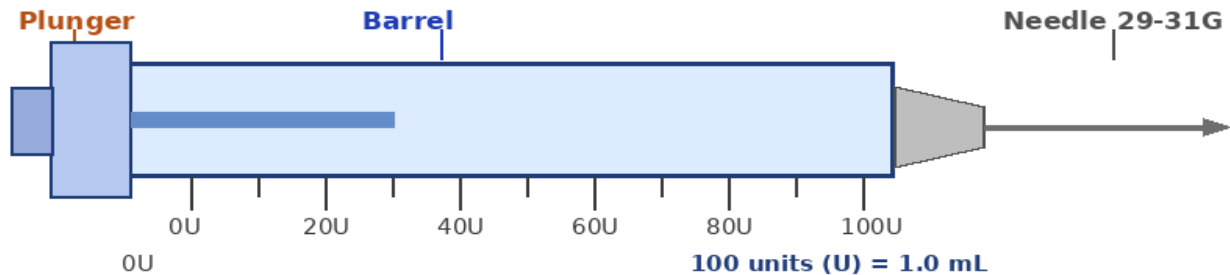
Reconstitution Details

Step	Action	Full Detail
1	Set temp to 20°C	Remove sterile water and NAD+ vial from fridge. Leave at room temperature 15-20 minutes. Cold mixing causes incomplete dissolution.
2	Sanitise cap	Wipe rubber stopper of BOTH vials with separate alcohol swabs. Let air-dry 15 seconds. Do not touch after wiping.
3	Draw 2mL water	Use a 3mL or 5mL syringe. Insert needle into sterile water vial, invert, pull back plunger to exactly 2mL. Remove air bubbles.
4	Inject along wall	Insert needle into NAD+ vial. Tilt so needle tip touches inside glass wall. Push SLOWLY — let water run down the wall onto the powder. Do NOT spray directly onto powder (causes foaming and degradation).
5	Swirl — no shaking	Gently roll vial between palms or swirl in circles. NAD+ dissolves in 30-60 seconds. Solution should be clear with faint yellow tint. DISCARD if cloudy, pink, or contains visible particles.

After reconstitution: Label vial with date and time. Store at 2-8°C. Use within **24 hours** — NAD+ degrades faster than most peptides once dissolved. Do NOT freeze reconstituted solution.

SECTION 3 — Dose Calculation

How to Read the Syringe (100U = 1.0 mL)



Each 10 units (U) = 0.10 mL. 100U = 1.0 mL full volume.

Dose Calculation Formula

$$\frac{\text{DOSE (mg)}}{\text{CONC. (mg/mL)}} = \text{VOLUME (mL)} \times 100 = \text{UNITS (U) on syringe}$$

Example — 500mg vial (250mg/mL): 250mg dose → $250/250 = 1.0\text{mL} = 100\text{U}$ | 100mg dose → $100/250 = 0.40\text{mL} = 40\text{U}$

Example — 1000mg vial (500mg/mL): 250mg dose → $250/500 = 0.50\text{mL} = 50\text{U}$ | 500mg dose → $500/500 = 1.0\text{mL} = 100\text{U}$

500mg Vial (250 mg/mL after 2mL water)

Dose (mg)	Volume (mL)	Units (U)	Research Context
100 mg	0.40 mL	40 U	Low / introductory dose
250 mg	1.00 mL	100 U	Standard dose — most protocols
500 mg	2.00 mL	200 U	Full vial — high dose protocol

1000mg Vial (500 mg/mL after 2mL water)

Dose (mg)	Volume (mL)	Units (U)	Research Context
250 mg	0.50 mL	50 U	Low / introductory dose
500 mg	1.00 mL	100 U	Standard dose — most protocols
750 mg	1.50 mL	150 U	High dose — advanced protocol
1000 mg	2.00 mL	200 U	Full vial — intensive protocol

Most common starting dose: 250mg. First-time users should always start at 250mg or less to assess tolerance before increasing.

SECTION 4 — Administration: IV Drip & SubQ Injection

Infusion Rate — The Most Critical Rule



Slower = Safer. Side effects are ALWAYS caused by infusion speed.

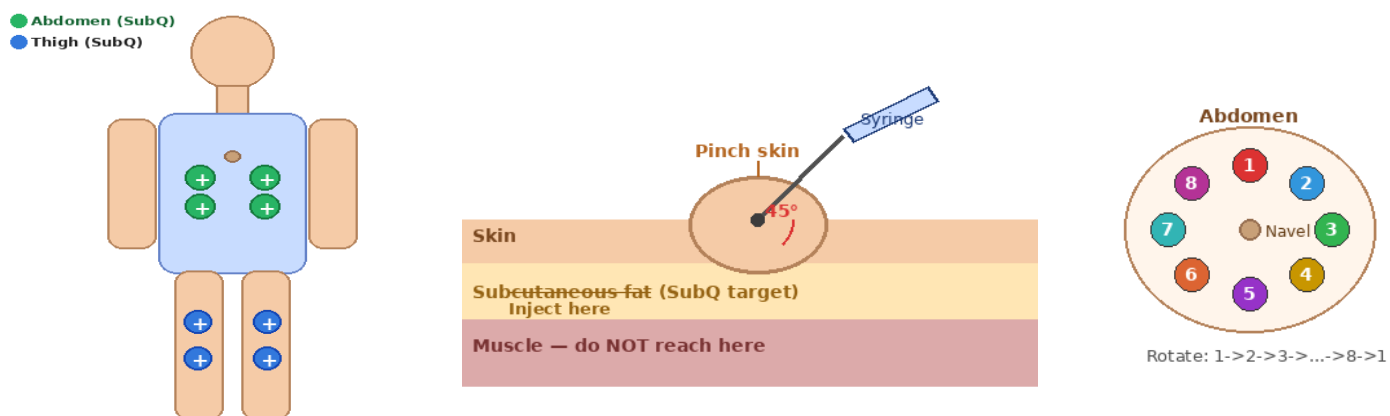
IV Drip Protocol (Recommended Method)

Step	Action	Detail
1	Prepare IV bag	Add reconstituted NAD+ to 100-250mL Normal Saline (NS 0.9%) IV bag. Mix gently. Cover bag with foil if light-sensitive.
2	Set drip rate	Set for 2-4 hour infusion. Use IV drip calculator: 250mL over 4hrs = ~62 drops/min (standard drip set).
3	Monitor first 15 min	Stay with patient. Watch for flushing, chest tightness, or nausea. These mean: SLOW DOWN the drip rate — do not stop entirely.
4	Adjust if side effects	If nausea or flushing occur: reduce drip rate by half. Side effects always resolve when rate is reduced.
5	Flush line after	After NAD+ completes, flush IV line with 20-30mL plain NS before removing or switching to another infusion.

Supplies needed for IV: IV bag (NS 0.9%, 100-250mL), IV line/tubing, IV cannula (20-22G), alcohol swabs, gloves, drip stand, tape.

SubQ Injection Method (Alternative)

If IV administration is not available, NAD+ can be given subcutaneously. SubQ absorption is slower but effective for maintenance dosing. Maximum 1mL per injection site — split larger doses across multiple sites.



Left: SubQ injection sites (green=abdomen, blue=thigh). Centre: Pinch skin, insert at 45°, push slowly. Right: Rotate 8 sites in order — never repeat same spot.

SECTION 5 — Dosing Protocol & Weekly Schedule

Recommended Daily Timing



Morning infusion is standard. A second evening dose is optional and only if prescribed.

Research Protocols by Goal

Goal	Dose	Frequency	Duration	Notes
General wellness & energy	250-500 mg	2x per week	4-8 weeks	Good starting protocol
Anti-aging & longevity	500 mg	2-3x per week	8-12 weeks	Maintain with 1x/week after
Cognitive enhancement	250-500 mg	2x per week	6-8 weeks	Can repeat cycles
Addiction recovery	500-1000 mg	Daily x 10 days	10 days	Clinical supervision required
Athletic recovery	250-500 mg	1-2x per week	Ongoing	Best taken day after training
Rest / Off period	—	—	4 weeks	Allow 4 weeks before new cycle

SECTION 6 — Side Effects, Safety & Storage

Supply Duration Calculator

Vial	Dose per session	Duration (2x/week)
500mg vial	250 mg (standard)	~1 week per vial
500mg vial	500 mg (full vial)	~3-4 days per vial
1000mg vial	250 mg (standard)	~2 weeks per vial
1000mg vial	500 mg (standard)	~1 week per vial

Note: Reconstituted NAD+ must be used within 24 hours. Store unconstituted powder at 2-8°C refrigerated.

Side Effects — Almost Always Rate-Related

NAD+ IV side effects are almost exclusively caused by infusing too fast. Slowing the drip rate resolves 95% of reactions within minutes.

Side Effect	Cause	Severity	Solution
Flushing / warmth	Too fast infusion	Mild	Slow drip rate immediately
Chest tightness	Too fast infusion	Mild-Mod	Slow drip — monitor closely
Nausea	Too fast infusion	Mild	Slow drip; small meal before
Headache	Rate or dehydration	Mild	Hydrate well before infusion
Palpitations	Too fast infusion	Mild	Slow drip immediately
Injection site redness	SubQ — normal	Mild	Rotate sites each time
Allergic reaction	Rare	Serious	STOP immediately — seek help

Golden rule for NAD+ IV: If any side effect occurs — slow the drip, do not stop it. Stopping abruptly and restarting can worsen symptoms. Simply reduce the rate and all effects resolve within 5-10 minutes.

When to STOP and Seek Medical Attention

- Severe chest pain or pressure that does not resolve after slowing drip — STOP and call emergency
- Allergic reaction: hives, difficulty breathing, or facial swelling — EMERGENCY
- Fever above 38°C or signs of infection at IV/injection site
- Solution is cloudy, discoloured, or contains particles — DISCARD immediately
- Vomiting that prevents staying hydrated — discontinue and rehydrate

Storage Summary

State	Temperature	Max Duration	Notes
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Lyophilized powder (unreconstituted)	2-8°C refrigerated	Per expiry date	Keep dry; do not freeze
Reconstituted solution	2-8°C refrigerated	24 hours ONLY	Label with time; degrade fast
In IV bag (with NS)	2-8°C or use immediately	Within 4 hours	Infuse same day — no storage
BAC/Sterile Water opened	2-8°C refrigerated	28 days	Label with open date

10 Golden Rules for NAD+ Administration

#	Rule
1	SLOW is SAFE — never rush a NAD+ infusion. 2-4 hours minimum.
2	Always start at 250mg or less for first-time infusions.
3	If any side effect occurs — slow the drip, do not stop it.
4	Use reconstituted NAD+ within 24 hours — it degrades quickly.
5	Store unreconstituted powder at 2-8°C refrigerated.
6	Hydrate well before and during infusion — drink 500mL water beforehand.
7	Never infuse cold solution — always bring to room temperature first.
8	For SubQ: use a new needle every injection, rotate sites 1 through 8.
9	Label every reconstituted vial with date and time prepared.
10	Discard any solution that is cloudy, pink, or contains particles.

DISCLAIMER: This guide is for informational and research reference purposes only. NAD+ is Research Use Only (RUO). IV administration should only be performed by or under the supervision of a qualified medical professional. SaiyanMed is not responsible for misuse of this product.

Provided by SaiyanMed | Research Use Only | NAD+ 500mg & 1000mg | Purity 99.8% — Endotoxin-Free — Heavy Metal-Free